

## Hyperkalemia

### ABIM Core Principles for Hyperkalemia Management (MEDICAL EMERGENCY):

1. **EKG Changes (Peaked T waves, wide QRS):** Step 1: Stabilize myocardium -> Give IV Calcium Gluconate.
2. **Step 2: Shift K+ into cells** -> Give IV Insulin with D50 and Albuterol.
3. **Step 3: Remove K+ from body** -> Use Furosemide, Potassium Binders (e.g., Kayexalate, Lokelma), and/or Dialysis.

## Hypokalemia

### ABIM Core Principles for Hypokalemia Management:

- **High-Yield Principle: CHECK MAGNESIUM.** Hypomagnesemia makes potassium repletion refractory. Replace Mg first.
- Oral KCl is preferred for mild cases. Use IV KCl for severe/symptomatic states (infuse slowly).

## Disorders of Phosphate and Magnesium

### Keywords / Typical Presentation

- **Hypophosphatemia:** Most often asymptomatic. Severe (<1.0 mg/dL) can cause muscle weakness and respiratory failure (diaphragm), and hemolysis. Key causes are Refeeding Syndrome, Diabetic Ketoacidosis (DKA), and Alcoholism.
- **Hyperphosphatemia:** Usually seen in End-Stage Renal Disease (ESRD). Can cause soft tissue calcification and symptoms of co-existing hypocalcemia.
- **Hypomagnesemia:** Seen in **Alcohol Use Disorder**, malabsorption, or with drugs (PPIs, Diuretics). Presents with **neuromuscular excitability** (tremor, tetany, seizures). Can cause refractory hypokalemia and hypocalcemia.
- **Hypermagnesemia:** Rare, primarily due to ESRD or excessive laxative/antacid use. Can cause **loss of deep tendon reflexes**, hypotension, and respiratory depression.

### ABIM Core Principles for Management:

- **Hypophosphatemia:** Mild is oral repletion. Severe/Symptomatic requires **IV Phosphate**. **CAUTION:** Rapid IV repletion risks **hypocalcemia** and metastatic calcification.
- **Hyperphosphatemia:** Management is a low-phosphate diet and Phosphate Binders (e.g., Calcium acetate, Sevelamer). Severe or symptomatic requires **Dialysis**.
- **Hypomagnesemia:** Mild is oral repletion. Severe/Symptomatic requires **IV Magnesium Sulfate**.